Soil Prediction System for Soil Health Classification

Abstract

This project presents a Soil Prediction System developed using the scikit-learn machine learning library to classify soil health and provide intelligent agricultural recommendations. The system analyzes soil composition data such as nitrogen, phosphorus, potassium, pH level, and moisture content to determine soil quality. Based on classification results, the model recommends crop rotation strategies and fertilization plans to improve soil productivity and sustainability.

Introduction

Soil health plays a critical role in agricultural productivity and sustainable farming. Traditional soil analysis methods often require manual interpretation and expert consultation. Machine learning techniques can automate this process by analyzing soil composition data and predicting soil health conditions.

Problem Statement

Farmers often lack accurate tools for evaluating soil health and determining the appropriate crop rotation or fertilizer strategy. Without scientific analysis, soil nutrients may become depleted, leading to lower crop yields and unsustainable farming practices.

Objectives

The primary objective of this project is to develop a machine learning model that classifies soil health based on soil nutrient composition and environmental parameters. The system also aims to generate recommendations for crop rotation and fertilization to improve soil fertility.

System Architecture

The system consists of modules for soil data collection, preprocessing, machine learning model training, soil health classification, and recommendation generation using the scikit-learn library.

Dataset Features

The dataset includes parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH value, moisture level, and organic matter content.

Model Implementation

The model is implemented in Python using scikit-learn. Data preprocessing includes handling missing values, normalization, and feature selection. A classification algorithm such as Random Forest or Decision Tree is trained on labeled soil datasets.

Model Evaluation

Model performance is evaluated using metrics such as accuracy, precision, recall, and confusion matrix.

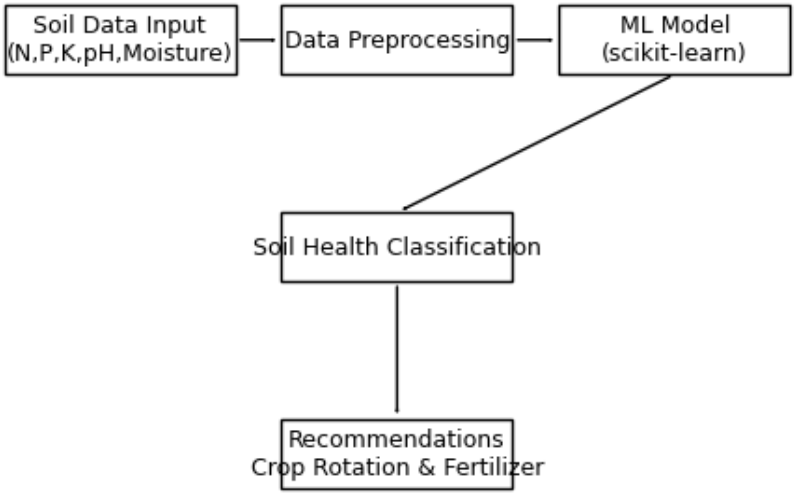
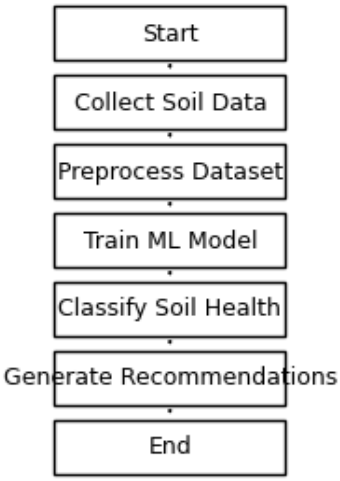
Applications

The Soil Prediction System can assist farmers, agricultural researchers, and government agencies in monitoring soil fertility and recommending sustainable farming practices.

Conclusion

The Soil Prediction System demonstrates the potential of machine learning in agriculture by providing intelligent soil health classification and recommendations.

System Architecture Diagram



Workflow Diagram